

Big Data analytics in railways lead to predictive analytics and make decisions based on huge amounts of data. These involve data collection, analysis, visualisation and decision-making for assets.

Estimation of the remaining useful life of an asset to ascertain the probability of its mission accomplishment is key to the success for any organisation. The railway domain can achieve data interconnection via the train bus where most railway sub-systems and their respective sensors are accessible for global optimisation.

With the increase in demand for more passenger rail services and greater volumes of cargo trains, use of data and its analysis will become a primary asset for the railway sector. Millions of data points captured from sensors on critical train components will help detect impending part defects, ensuring maintenance before a defect occurs. This will improve reliability of the system infrastructure for many years by remote monitoring of location and condition of all vehicles.

More technologies to meet needs of the railways

Researchers at Siemens Corporate Technology have developed a technique known as frequency-selective coating of window panes to solve the problem of low mobile Internet signals. These panes are provided with a transparent electrically-conductive layer consisting of metals or metal oxides. Metallic coating of the windows is vaporised along lines in a special structure by a laser to pass certain frequency ranges unobstructed.

The VR lab at GE Transportation's John F. Welch Technology Centre, Bengaluru, facilitated a 3D VR environment for inspection. It tracks the motion of inspectors through specially-designed wearable goggles and infrared (IR) cameras placed in the room. The VR controller allows users to interact with individual components. The environment helps teams at different locations collaborate and review products that are under develop-

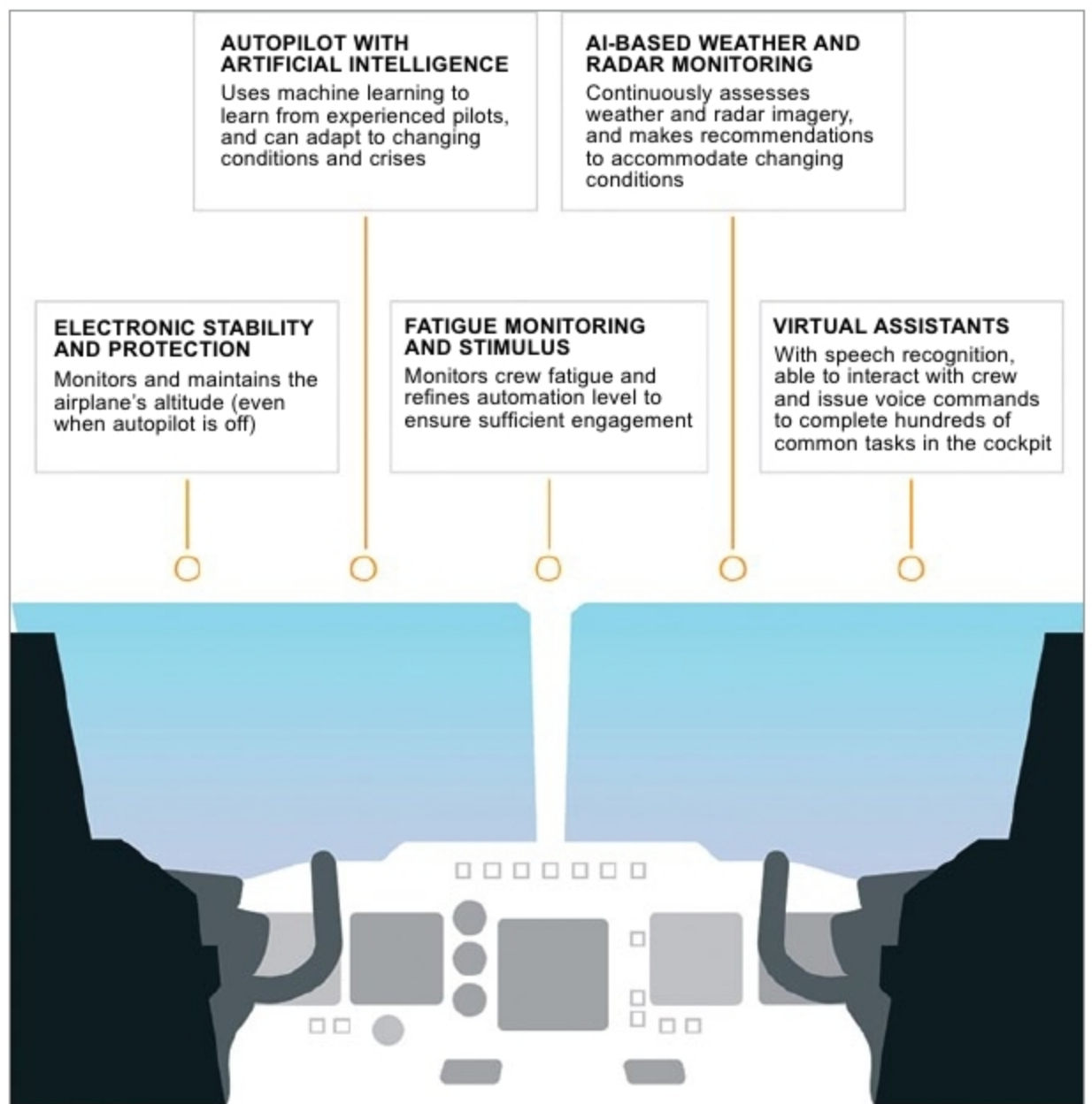


Fig. 3: Possible automation of the cockpit (Credit: www.oliverwyman.com)

ment. The lab can also be used as a training ground for service engineers who work on products in the field.

Locotrol distributed power system is a control and communication system that enables coordinated braking and traction power distribution between lead and remote engines for faster stopping times and shorter stopping distances. Locovision system through superior image quality cameras and real-time data processing monitors wayside assets, measures track gauges and detects intruders. It stores all information in a hardware infrastructure to help avoid major asset repairs and fines. The rail integrity monitor employs innovative technology with mounted sensors under the locomotive, continuously testing rail integrity in real time.

Automatic train control systems continuously monitor all train movements to provide fail-safe signalling. Operation of railways is centrally-monitored and controlled through

operations control systems. Supervision systems, such as CCTVs or emergency telephones, also contribute to safety and enhanced comfort.

Signalling systems and railway automation solutions are crucial to detect and signal whether line sections are clear or occupied. There are three grades of automation and train control systems, namely, partially automated (supervised by a driver), highly automated (reduced driver supervision) and fully automated (the system is responsible).

Airways technologies and automation

The year 2019 looks to be the era of digital transformation with technology becoming a part of everyday life within all industries, including air transport industry. The airline industry is constantly evolving and adapting to new consumer habits, expectations and requirements for better safety and security. Technologies and trends that we expect